

An early implementation of a DAVIC V1.0 system — use of dynamic connections for interactive multimedia services

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Advances in networking technology and the provision of greater wide area bandwidths are the catalysts for the emergence of new broadband services in both the business and residential market. This is in turn producing new, more complex, business models involving service consumers, content providers, service brokers and network connectivity providers. Such business models require the underlying technology to enable sophisticated network signalling to set up, monitor and tear down connections reliably and in real time.

This paper describes the evolution of a broadband call control demonstrator believed to be the first of its kind in the world to provide a practical realisation of the signalling required in such business models. It focuses on signalling and switch control and has developed into an end-to-end implementation of a Digital Audio Visual Council (DAVIC) V1.0 system.

The evolution of the switch control and signalling architecture is described, showing where existing standards have been implemented to provide basic connectivity across an ATM network. Additional capabilities are built upon this platform through the adoption of emerging signalling concepts, such as session control, and application control using distributed object-oriented principles, for the access and control of a video-on-demand service.

To further understand the evolutionary path for emerging consumer systems, the paper illustrates how work was undertaken to separate set-top box (STB) functionality into a set-top unit (STU) and a network interface unit (NIU), allowing easier access to future broadband network capabilities.

1. Introduction

To enable the introduction and operation of world-class basic and advanced broadband services, BT has developed expertise in the provision of on-demand broadband connections, with their high bandwidth and guaranteed qualities of service. Work has been under way for several years investigating the issues associated with such on-demand connections.

A significant part of this work involved the development of a demonstrator platform, on which basic signalling and call-control concepts could be explored. This has now evolved to represent an end-to-end implementation of a DAVIC V1.0 system, believed to be the first of its kind in the world.

1.1 Objectives

The objective of this paper is to describe how, through the use of high-level application programming interfaces (APIs) and an evolutionary architecture, the development was able to progress through a number of discrete phases. At each phase a specific set of more advanced signalling

protocols could be explored, all leading towards the provision of a video-on-demand service. Finally, it describes how the platform is being continually developed to retain a leading edge in this field.

1.2 Background

When this work started, the majority of asynchronous transfer mode (ATM) switch vendors only supported a permanent virtual circuit (PVC) capability, where connections are configured through a management interface. Some ATM switch vendors did support a switched virtual circuit (SVC) capability, although this was based on proprietary implementations.

Standards were available, namely the ITU-T Access Signalling Recommendation (Q.2931) [1] and the ATM Forum User Network Interface Signalling Recommendation (ATM-F V3.0) [2]. Software implementations did exist, although not integrated with any ATM switch vendors' products. These could be purchased at a premium.

Faced with the objective of understanding the issues associated with broadband connections, it was decided that, for a comprehensive understanding and for future flexibility, the best solution was for the team to develop its own implementation of the signalling standards.

However, to integrate this software with an ATM switch, the team also required comprehensive access to an ATM switch vendor's control software. This was not forthcoming either, and for similar reasons the team opted to develop its own switch control software.

Switch control

The switch control software was developed on a separate control terminal. Signalling, call control, address server, connection admission control (CAC), connection control and resource manager functionality were implemented. The ATM switch was controlled by a simple network management protocol (SNMP)-based interface that allowed connections to be configured across the ATM switch at the request of the control software.

If support for different vendors' ATM switches is required the flexibility of the switch control software means that only the interface between the connection control and the ATM switch would require modification.

Point-to-point capability

The point-to-point connection capability was provided by the ITU-T Q.2931 signalling recommendation. However, due to the complexity of this recommendation it was agreed that only a sub-set of the full standard was required. Parts of the recommendation covering timers and detailed release causes were omitted.

The initial demonstration supported a simple point-to-point connection capability using the separate control terminal to terminate the messages sent from the user terminals. Based on the connections requested the SNMP-based interface was then used to configure the ATM switch.

This provided a simple means of supporting on-demand connections (see Fig 1).

Point-to-multipoint capability

The basic point-to-point capability was developed to support a point-to-multipoint connection capability (see Fig 2). This was achieved using the ITU Q.2971 signalling

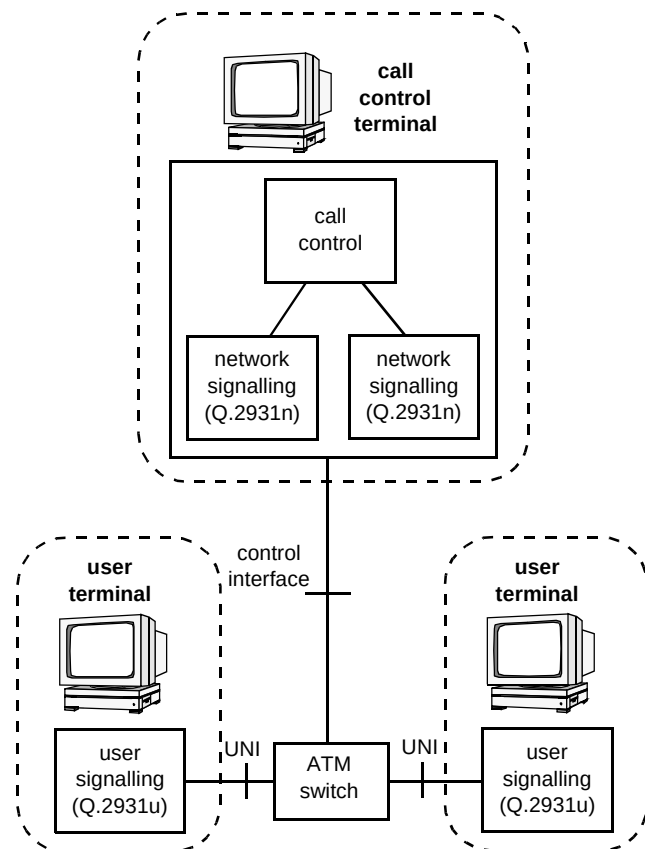


Fig 1 Initial demonstrator architecture — showing basic connectivity.

recommendation [3]. This was further extended with the use of the ATM Forum leaf initiated join (LIJ) procedures [4].

Proxy signalling agent

The next stage of development introduced the role of a proxy signalling agent (PSA). This is a process whereby the agent can signal on behalf of another network element which does not itself support the necessary signalling capability (see Fig 3).

The root terminal signals on behalf of a network element (a video server) which does not support signalling. This enables a point-to-multipoint connection to be established between the video server and a number of terminals, in effect representing a video distribution application.

A more recent phase of the demonstrator extended this concept by using a PSA to signal on behalf of a user's terminal or device. This uses lightweight signalling to notify the PSA that a connection is required, and for the PSA to notify the user's terminal or device of relevant connection information, e.g. virtual path identifier (VPI)/virtual channel identifier (VCI). This is a more realistic scenario, as

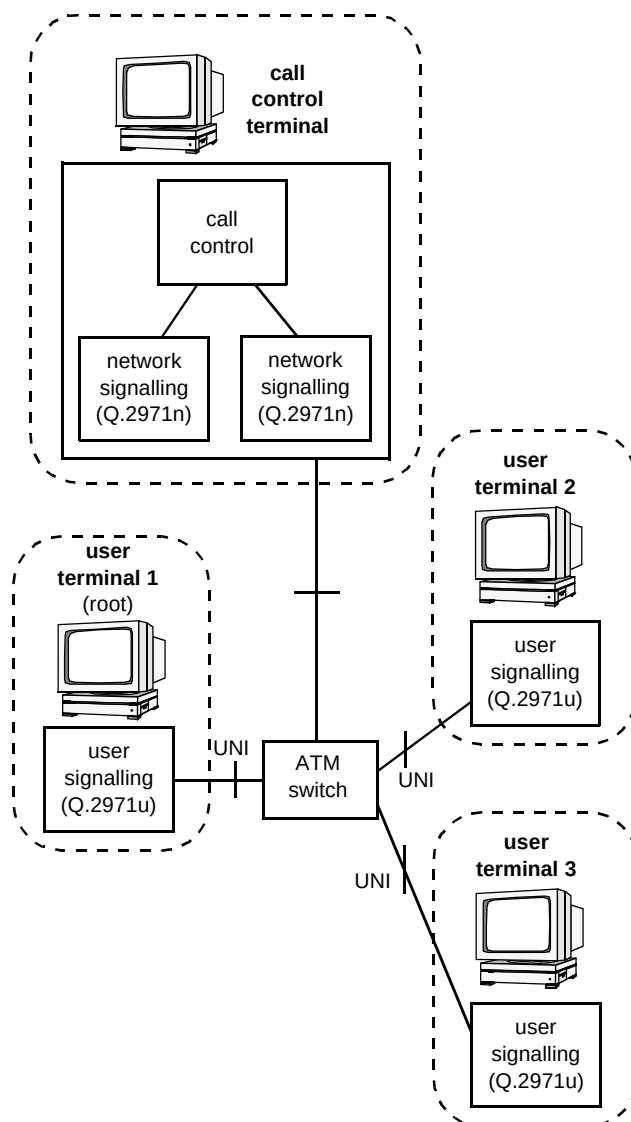


Fig 2 Point-to-multipoint demonstration.

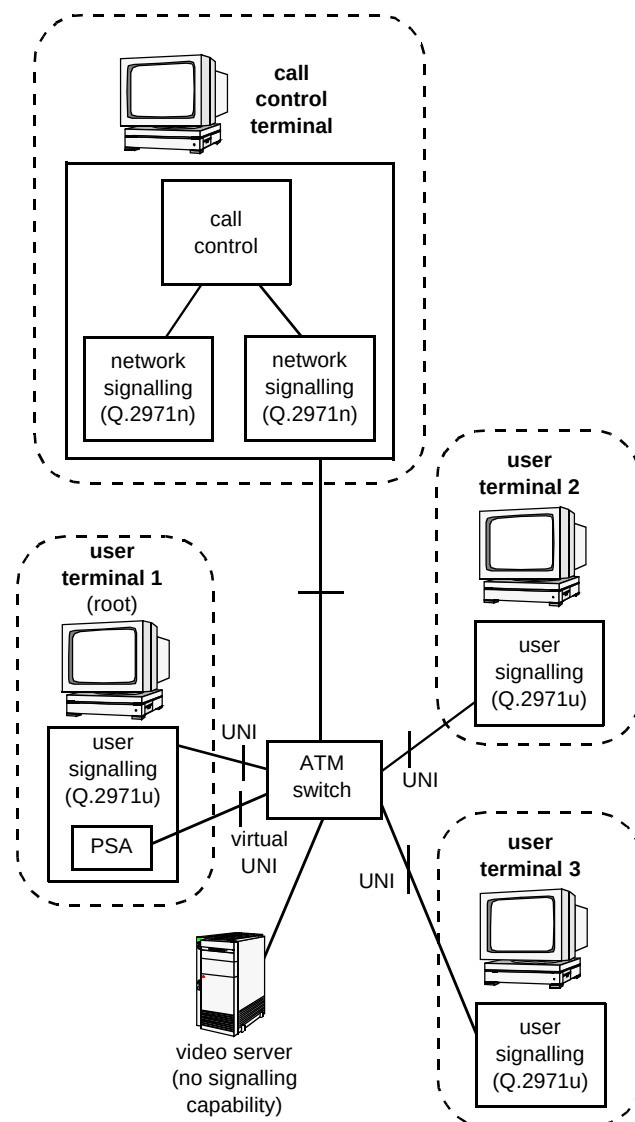


Fig 3 Basic proxy signalling — on behalf of a network element.

it means that full signalling (Q.2931) is no longer required on the customer's equipment, thereby reducing its cost. Additionally, the PSA could be 'shared' between many users, giving a more cost-effective solution.

2. Demonstrator platform

From the initial focus on call control issues described in section 1.2 the scope of the demonstration platform widened to cover more complex signalling mechanisms. This resulted in the requirement for a flexible demonstrator platform architecture and additional signalling functionality, in the form of session control.

This section describes the evolution of that enhanced network signalling functionality and its role within a future broadband network.

2.1 Business domains/roles

A long-term architecture was required to ensure a structured evolution capability for the demonstrator platform. TINA-C was the key consensus-forming body in this area and their future network architecture was adopted as the basis for evolution of the demonstrator platform.

TINA-C has identified five business domains/roles as part of the process of defining a future network architecture — service consumer system (SCS), network connectivity provider, retailer, broker and service provider system (SPS).

ITU-T SG11 has also adopted these domains and added a sixth, management provider, as part of their baseline text for CS-3. The definitions given in the following text are from the CS-3 baseline text:

- 'service consumer system' enrolls for and uses services,
- 'network connectivity provider' supplies the communications services that consist of calls and connections,
- 'retailer' provides the contact point for the consumer to arrange for, and the contact point for the service provider to offer, services which employ communications services as the delivery mechanism,
- 'broker' provides location information that the business domains use to locate other entities for the purpose of establishing communications,
- 'service provider system' delivers a variety of services which employ communications services as the delivery mechanism,
- 'management provider' is also defined by the ITU-T as a separate domain which handles the management of the other domains — since the management provider domain is not key to this paper, its presence is not considered further.

Figure 4 shows the domains/roles in the ITU-T CS-3 business model. The lines between these domains/roles are the inter-domain/role reference points.

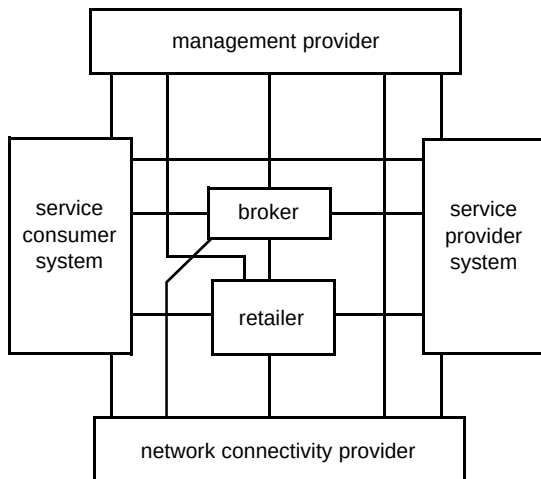


Fig 4 ITU-T CS-3 business model.

Figure 5 shows how the business model in Fig 4 has been applied to the demonstrator platform. The broker and management provider roles have not yet been implemented.

2.2 Session control

After the switch control and call/connection control functionality had been implemented, the next key area to consider was session control. This is a new concept developed from the DAVIC consensus forming body who use session control as the basis of an interactive multimedia session, where the session consisted of calls/connections delivering video, audio, etc. The session

control described in the DAVIC V1.0 specification [5] was based on digital storage media command and control (DSM-CC) [6]. The concepts of session control have since been adopted in ITU-T CS-3.

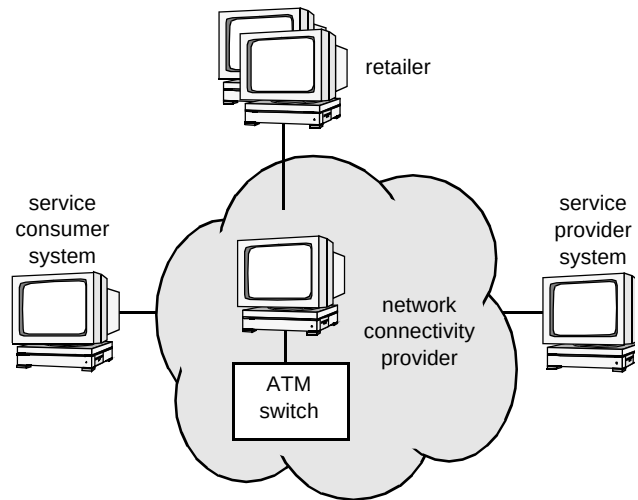


Fig 5 Application of business model to demonstrator platform architecture.

The DAVIC delivery system ties the session and call/connection control concepts closely together. Applying the TINA-C architecture, the session control functionality was located in the retailer. The network connectivity provider retained the call/connection control functionality. This provided a more flexible solution that would allow different operators to take retailer and network connectivity provider roles (see Fig 6).

As had been implemented in the past with the call/connection control, a high-level API to the session control level was defined, for use by higher-layer applications.

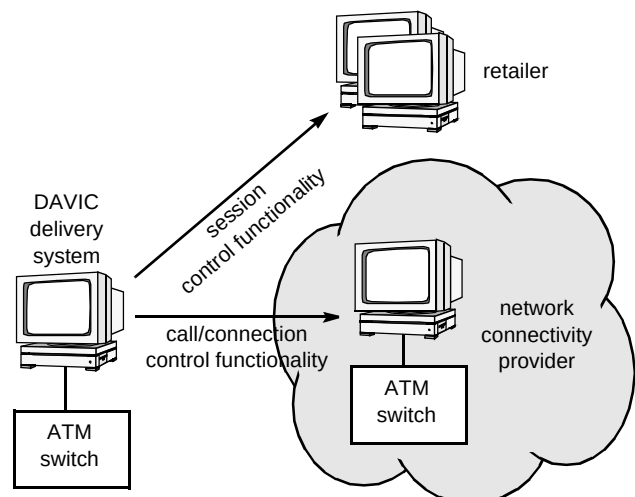


Fig 6 DAVIC delivery system mapped to the demonstrator platform architecture.

2.3 Demonstrator architecture and protocols

Figure 7 shows the demonstrator platform that was developed through implementation of the business domains/roles concept and the implementation of session control.

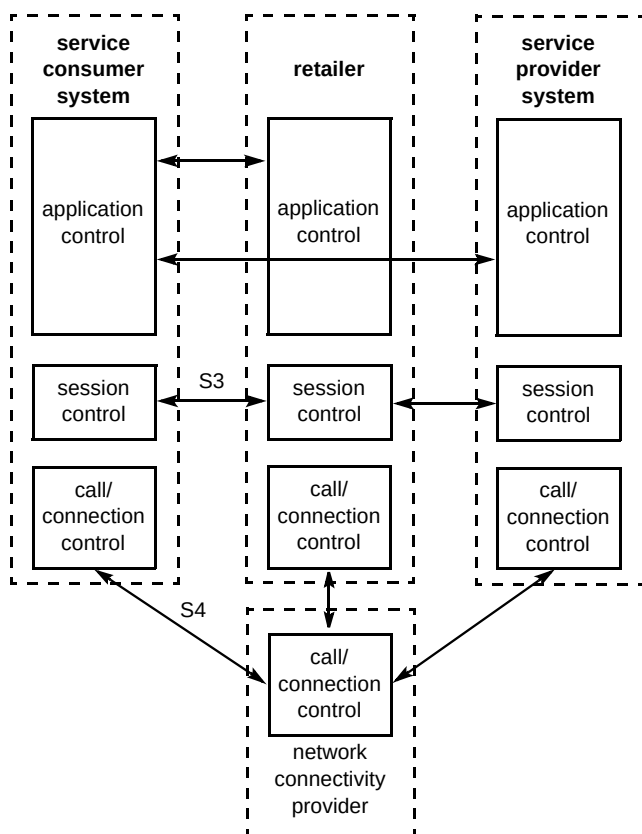


Fig 7 The demonstrator platform architecture and protocols.

The session control uses the call/connection control protocols to configure connections for the higher-layer application. Figure 7 also introduces the concept of S-flows which come from DAVIC and has been used throughout the development work.

The S3 flow is a bi-directional flow used for the exchange of session information. The protocol specific to the S3 flow is the International Standardisation Organisation (ISO) Motion Pictures Expert Group (MPEG) session control protocol, DSM-CC user-network (U-N).

The S4 flow is a bi-directional flow supporting call/connection control and resource control functions. The call/connection control functions consist of the capabilities to establish calls and connections in a B-ISDN network. The protocol specific to the S4 flow is the ITU-T call/connection control protocol, Q.2931.

3. Platform development

This section describes the enhancements built upon the platform described earlier in the paper, making use of the session control API to assist development and integration phases.

The first phase of development work built an application control and presentation layer on top of the session control, call/connection control and switch control functionality (see Fig 8).

3.1 Application control

The application control layer developed on the platform resides on top of the session control layer, with correspondent functionality situated within the SCS, retailer and SPS. The choice of protocol used for this layer has come from DAVIC V1.0 and is based on the ISO MPEG DSM-CC user-user (U-U) recommendations. DAVIC also refers to the application control layer as the S2 level.

The SCS application is responsible for presenting the user with a browsing interface to the S2 level service domain. As such the user can choose to attach a session to a service gateway (retailer or SPS), navigate directory structures to select services (movies on demand, Internet access), retrieve files (executable, graphical or text), and exert control over MPEG streams (play, pause etc). The graphical user interface (GUI) chosen for the demonstrator system was a hypertext mark-up language (HTML)-based Web browser. Obviously Web browsers do not communicate using DAVIC protocols, namely DSM-CC U-U, but use the hypertext transfer protocol (HTTP) between browser and webserver. Therefore the end-user application is a combination of webserver software with the ability to translate from HTTP requests to DSM-CC U-U method invocations and then back again to HTML, so that the user can continue to navigate the application. The SCS application is also responsible for controlling the MPEG decode hardware within the end-user system in response to S2 stream control remote procedure calls. This is discussed further in section 3.2.

DSM-C U-U

Figure 9 positions the DSM-CC U-U protocol in the demonstrator platform software architecture.

The three software objects that are primarily concerned with the use of the DSM-CC U-U protocol are the application object, the object server and the naming service.

When a user wishes to attach to a retailer or an SPS they must first of all select that system from a local menu. The presentation layer conveys this selection down to the application object that translates it into an end-system

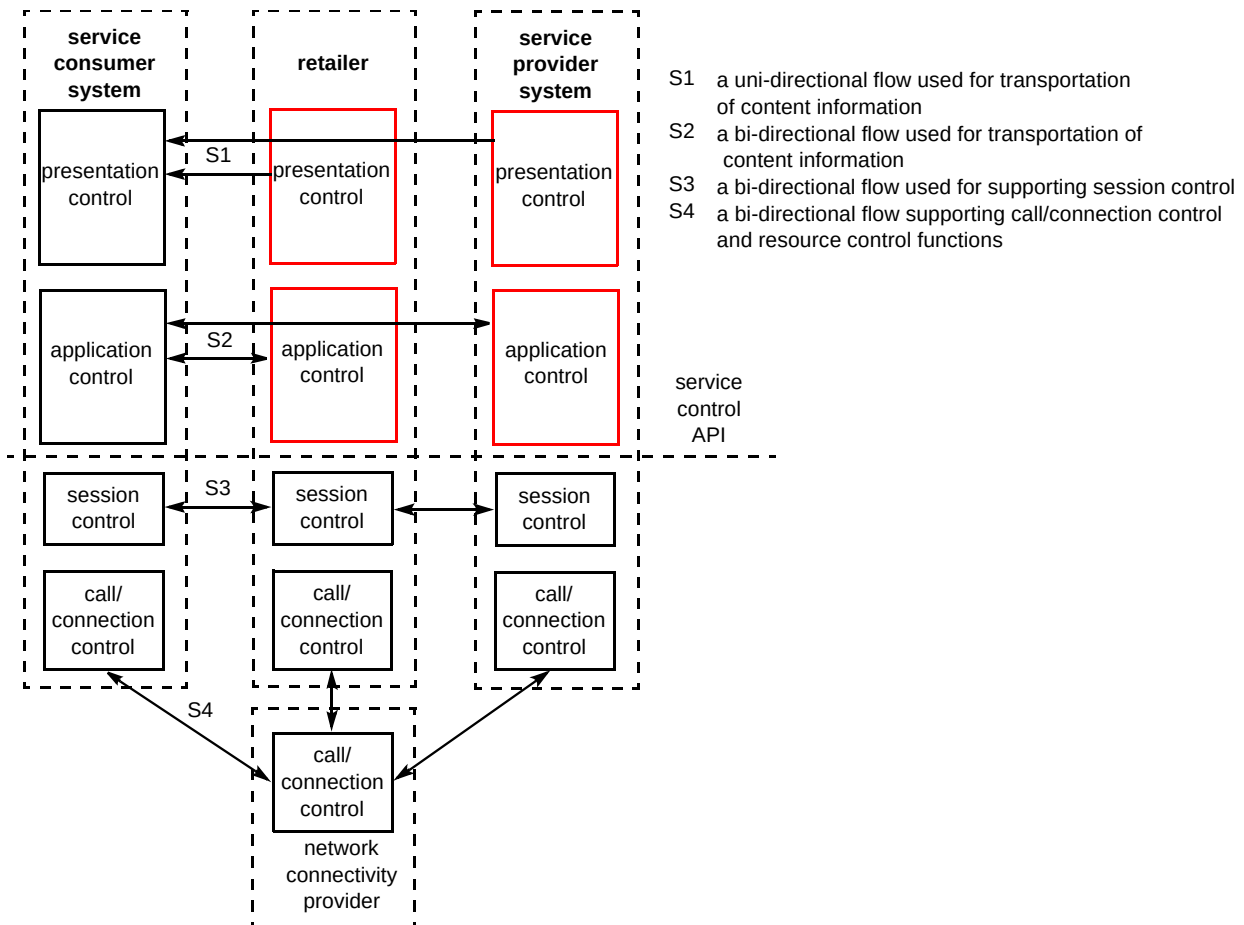


Fig 8 Platform development — phase 1

address. There are two scenarios by which this address could have been acquired:

- the user obtained it externally when registering with a retailer,
- the user obtained it on-line via a known retailer or service provider.

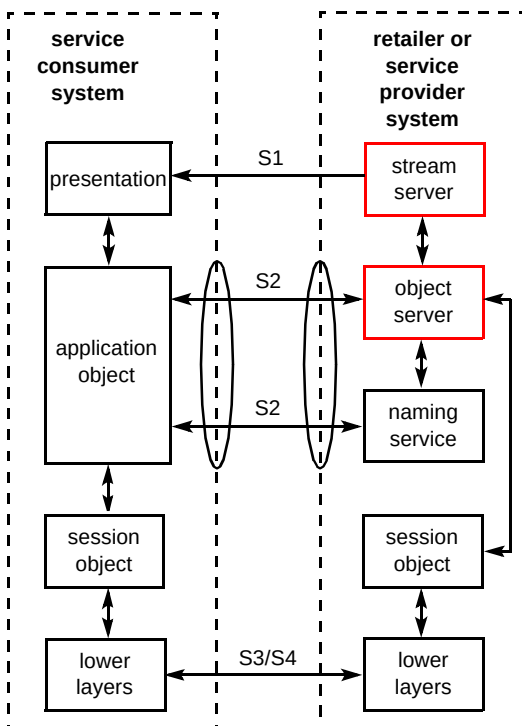


Fig 9 Position of the S2 protocol layer.

What the application object now needs is an object reference for the naming service of the system that the user wishes to browse. The naming service is the place where the object references for all of the services on the remote system are stored. This store is a directory hierarchy. Object references are used to create proxy objects for the application object which are used to communicate with the remote objects in the object server. Proxy objects make it seem to the application that the remote object is actually executing locally.

The object reference for the naming service is obtained via the two session objects, one in each system and the lower layers. The reference is retrieved from the object server where it was placed by the naming service. It is the only object reference that is not stored in the naming service.

Once the application object has the object reference, it uses it to create a proxy object for the naming service. The

user is then able to browse the naming service and therefore the content on the remote system.

Functions used to browse the naming service are also able to return the object reference held there across S2. This reference can then be used to create a proxy object for the remote service. A service can then be treated as if it resided on the user's machine.

In Fig 9 a stream server is shown. Each stream on the server represents a movie title. Streams are controlled by a stream object on the object server and therefore a user is able, via a proxy object on their own machine, to take control of the movie being transmitted to them on S1.

Common Object Request Broker Architecture (CORBA)

CORBA is used as an architecture for access to objects on the retailer and SPSs. Both retailer and SPSs can be thought of as object request brokers. They offer CORBA-compliant access to their objects in the form of a naming service. Furthermore, interfaces to objects across S2 are defined in interface definition language (IDL) and implemented by Orbix in a CORBA-compliant manner.

The Internet inter-orb protocol (IIOP) is used to transfer the initial naming service object reference from the object server (SPS/retailer) to the application object (SCS).

3.2 Presentation

The presentation layer developed on the platform sits on top of the application control layer, with correspondent functionality situated within the SCS, retailer and SPS. The choice of protocol used for this layer has come from DAVIC V1.0 and is based on ISO MPEG. DAVIC also refer to the presentation layer as the S1 level.

The presentation layer within the demonstrator is responsible for providing a GUI to the end user. This GUI comprises two parts — the browsing interface and the presentation interface for MPEG streams.

For the browsing interface, a commonly used Web browser was chosen. This was because Web browsers are a familiar paradigm and also have extremely powerful multimedia capabilities. Hence, the end user is presented with 'clickable' image and hypertext selections enabling navigation of the chosen service provider. Selection of, for instance, an action movies directory results in the invocation of the DSM-CC U-U directory resolve, which returns the contents of the directory for further browsing. Selection, and thus resolving, a stream object returns a set of stream control buttons to exert control over the stream.

MPEG

120 The S1 flow within the demonstrator has been implemented using the MPEG standard. In particular,

MPEG-1 system streams (SYS) have been chosen in preference to the DAVIC specified MPEG-2 transport streams (TS). This choice was made necessary because of the lack of mature, compatible MPEG-2 streaming and decode hardware during the hardware selection phase of the demonstrator build. This choice inhibits the transportation of private data across the S1 layer, which would otherwise have been carried within the private data part of the MPEG-2 TS.

For performance reasons the MPEG-1 SYS stream is mapped on to the ATM layer directly through ATM adaptation layer 5 (AAL5).

The MPEG content chosen for the demonstrator was encoded using a Minerva Compressionist MPEG encoding suite. The quality of the original material is highly important, therefore only Digi-Beta or Betacam SP quality material was used. Encoding rates chosen were between 1.6 Mbit/s and 3 Mbit/s video rate plus 224 kbit/s audio stream containing 44.1 kHz encoded stereo. Both audio and video streams were multiplexed on to a single MPEG SYS stream at rates of 1.89 Mbit/s and 3.24 Mbit/s respectively. With all MPEG material, better than VHS quality playback was observed. To quantify this statement, a video bit rate of 1.5 Mbit/s is quoted as being equivalent to VHS quality. This, of course, is highly dependent on the nature of the video material encoded.

Stream server

The demonstrator contains a single stream server, which is shared, over separate socket interfaces, by both the retailer and SPS. DSM-CC U-U stream remote procedure calls within the end-user application lead to appropriate RPCs between retailer/SPS and stream server (see Fig 10).

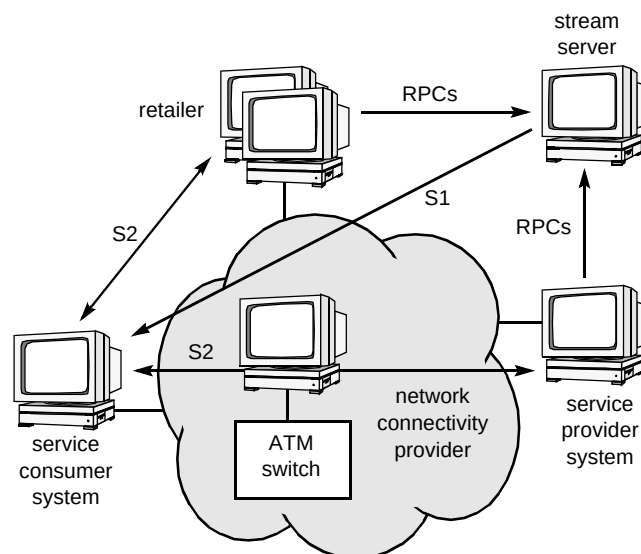


Fig 10 The demonstrator platform architecture (with stream server).

The stream server chosen is a Sun MediaCenter 5. This unit comprises a Sun workstation with operating system modifications specifically for loading and streaming real-time video, and a SCSI disk pack on which the MPEG streams are stored. The stream server offers a defined RPC interface over which a number of simultaneous streams may be controlled. The functionality of this RPC interface has been mapped on to the DSM-CC U-U stream interface under the control of a stream object within the retailer or SPS.

For the purposes of the current demonstration, streams are placed on to a predefined VCI.

MPEG player

In order to decode high-quality, real-time MPEG streams, the end-user terminal contains MPEG decode hardware. The hardware chosen for this purpose is the VideoPlex board, manufactured by Optibase. The VideoPlex software development kit was used to write the MPEG decode control software required by the end-user application.

This control software is controlled by the end-user application over a socket interface within the end-user terminal. When the user selects the play button associated with a stream, this instructs the MPEG card to read MPEG data directly from the ATM interface card as it becomes available. Meanwhile the stream server will be streaming the requested MPEG stream over a predefined VCI to the ATM interface card within the end-user terminal. Thus the play-back of the MPEG stream will commence.

The S-Video output from the MPEG decode hardware within the end-user terminal was connected to the video overlay card. A composite video output was connected to the television monitor.

The second phase of development work split the functionality within the SCS into set top unit (STU) and network interface unit (NIU) functionality (see Fig 11). The STU in phase 2 also supported no S4 signalling, requiring a PSA to be implemented in the retailer. The remaining phase 1 architecture was retained.

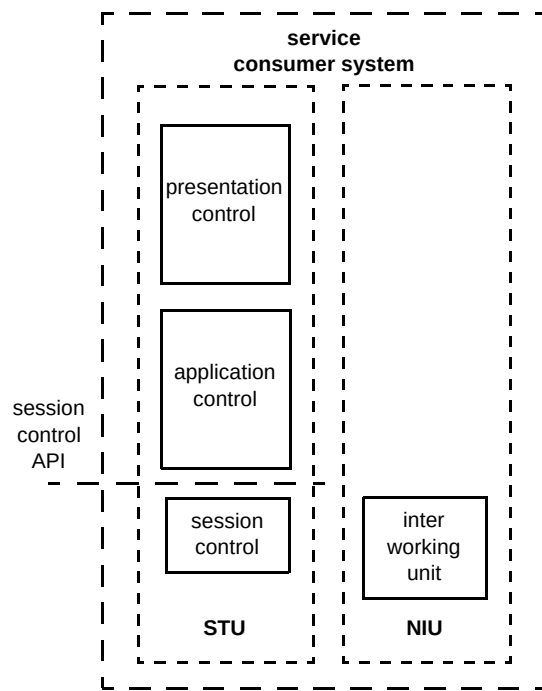


Fig 11 Platform development — phase 2.

3.3 Service consumer system (SCS)

The reason for this split was to investigate the feasibility of using IEEE1394 as the physical interface between an STU and a NIU and to validate the A0 (DAVIC-defined interface between STU and NIU — see Fig 12) local messaging protocols as specified in DAVIC. The IEEE1394 standard is rapidly being adopted as a high-speed digital interface on high-end digital consumer electronics such as camcorders and emerging set-top units.

This work would verify that STUs with an IEEE1394 interface would support connectivity (via an NIU) to future broadband telco networks (Fig 12).

The use of an external A0 interface between the STU and the NIU allows for portability across different delivery systems, e.g. broadband, satellite, etc, where the STU is designed to be network independent, with the NIU containing the access network specific functions.

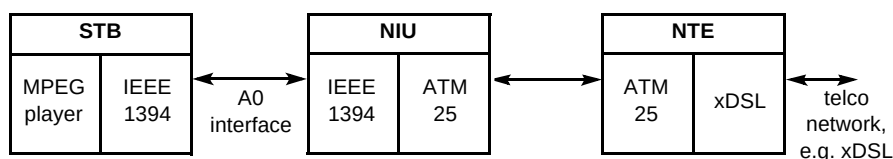


Fig 12 Phase 2 functional model.

STU and NIU functionality

The STU and NIU were modelled on two PCs and were heavily dependent on the code that had been developed for the SCS in phase 1 of the demonstrator. Essentially the S1 and S2 layers remained the same, with the main changes required at the S3 layer. No S4 layer was supported in the STU.

It was assumed that the NIU does not support ITU-T Q.2931 (S4) signalling, thus requiring proxy signalling functionality. In order to support the proxy connection, proxy signalling functionality had to be implemented on the retailer. It should be noted that during the initialisation process the STU informed the retailer that a PSA was required. A simplified functional model is shown in Fig 13 (the S1 and S2 layers are not shown).

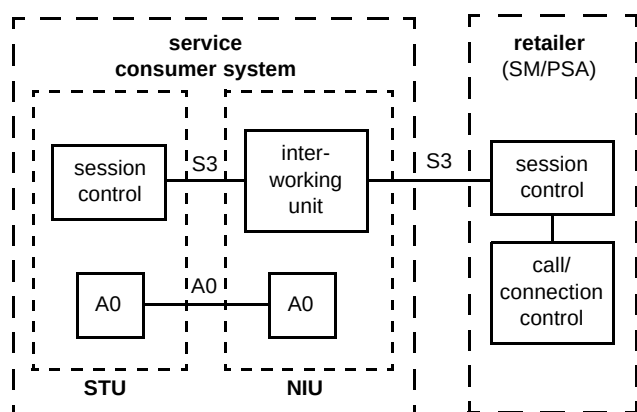


Fig 13 Phase 2 functional model.

Figure 13 also shows that local communication between the STU and the NIU is achieved by A0 messaging (defined in DAVIC) across the IEEE1394 physical interface. The other key control flow at the STU is session control (S3 flow), with the connection control (S4 flow) only required at the retailer (SM/PSA).

In this phase of the demonstrator it was assumed that the retailer would inform the STU of the VPI/VCI values for S1 and S2. No process existed by which the STU could request a VPI/VCI value from the retailer.

A0 interface and protocols

Three logical data buses were implemented on the A0 interface (as defined by DAVIC V1.0):

- the high-speed downstream data bus — up to 51.84 Mbit/s (carried S1 flow),
- the bi-directional cell bus — up to 2 Mbit/s (carried S2, S3 flows),

- the local control data bus — up to 100 kbit/s (carried DAVIC A0 messages, a minimum subset of messages being implemented for initialisation and call control across A0).

Figure 14 shows the protocol stacks which were implemented for phase 2 of the demonstrator.

The DSM-CC U-N (S3) messages were transferred directly over IEEE1394 between the STU and the NIU. The NIU had an interworking unit (IWU) which was responsible for reading the DSM-CC U-N messages from the ATM card and writing them to an isochronous channel (see below) for transport over the IEEE1394 interface.

A TCP/IP connection was then set up between the NIU and the delivery system. It should be noted that for delivery of the S1 content, the IWU was responsible for reading the MPEG data above the AAL5 layer and transporting these cells in an isochronous connection across the IEEE1394 bus.

IEEE1394 developer tool-kit

The STU and NIU both contained a 100-Mbit/s IEEE1394 PCI card (from Texas Instruments) with a Skipstone SerialSoft IEEE1394 developer tool-kit. The IEEE1394 serial bus is a high-speed digital interface for the transport of data between PCs and consumer electronic products. The tool-kit provided two transport modes:

- **asynchronous**, where the address of the sender and receiver is transmitted followed by the packet data — the asynchronous messaging was used on the demonstrator for transport of A0 messages between the NIU and STU (via A0 local control data bus),
- **isochronous**, where the sender requests an isochronous channel with a specific data bandwidth — two isochronous channels were used, one for the transport of the S1 flow (MPEG flow via high-speed downstream data bus) and the other for transport of S2 and S3 flows (DSM-CC flows via bi-directional cell bus).

The Skipstone SerialSoft layer provided an API for applications to perform asynchronous read/write transactions, as well as isochronous talk and listen transactions.

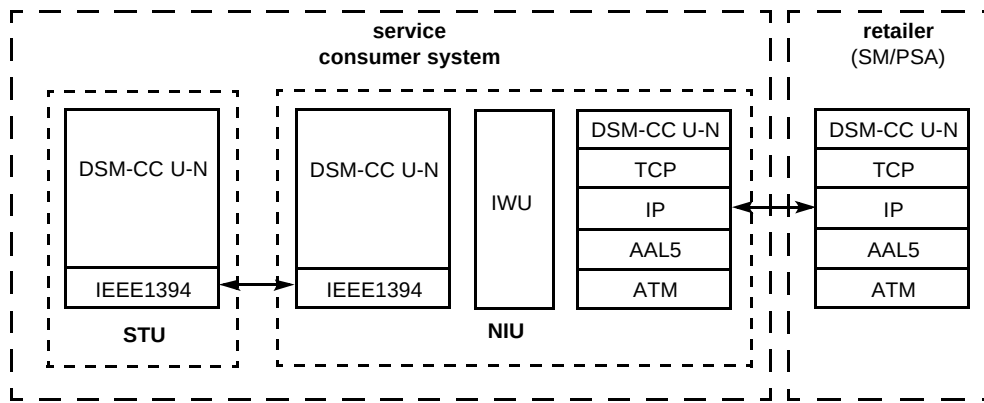


Fig 14 Phase 2 protocols.

STU implementation

The STU was modelled on a PC using a Skipstone IEEE1394 PCI card and the MPEG card which had previously been used in the SCS in phase 1 of the demonstrator platform.

The STU supported proxy signalling via the S3 flow using DSM-CC U-N, with a capability to establish and release sessions (e.g. ClientSessionSetUpRequest as in Fig 15). These messages were transported on a bi-directional cell bus via an isochronous IEEE1394 channel. The cell bus also supported DSM-CC U-U messages for signalling at the S2 layer.

The STU was also capable of accepting an incoming MPEG transport stream from the NIU via a separate isochronous IEEE1394 channel.

The STU supported a local control bus for communication with the NIU via asynchronous A0 messages across the IEEE1394 connection.

An example of the communication flows for establishing a session is also shown in Fig 15 (it should be noted that this assumes an initialisation stage has taken place between the STU and NIU).

NIU implementation

The NIU was modelled on a PC which contained a Skipstone IEEE1394 PCI card and an ATM25 PCI card.

The NIU effectively acted as a bridge by terminating the IEEE1394 physical layer and mapping the incoming or outgoing packets to the ATM layer. The NIU passed the VPI/VCI value to the STU, which then completed the

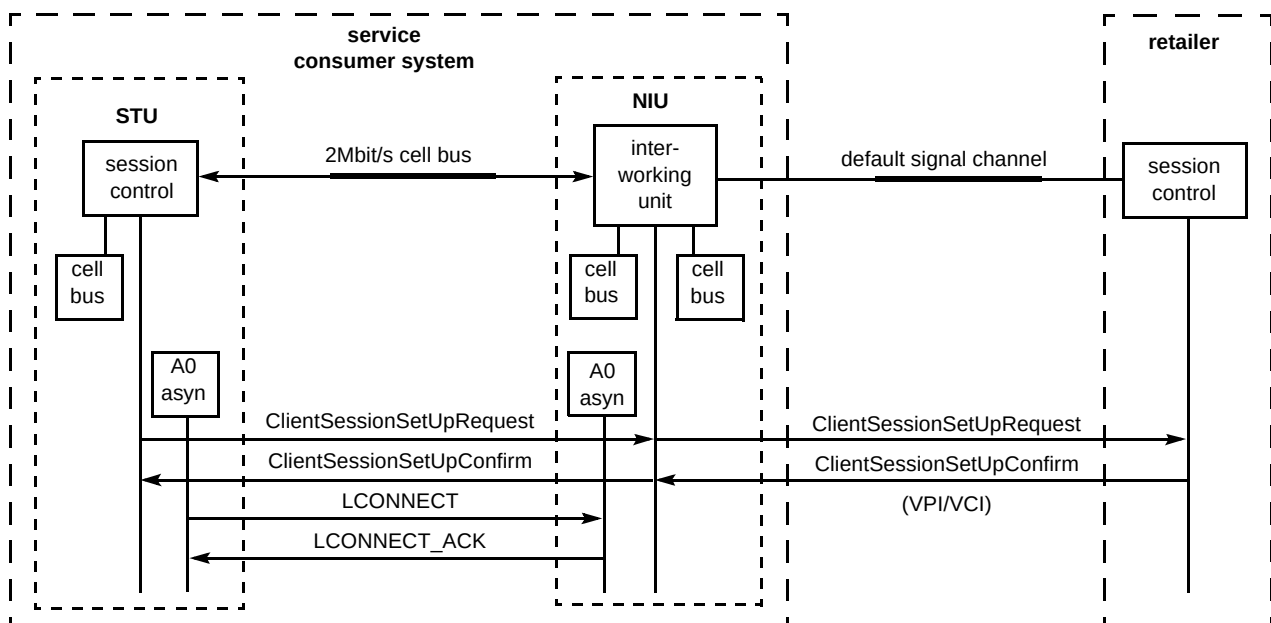


Fig 15 Session set-up flow example.

connection and allocated the correct IEEE1394 isochronous channel between the STU and the NIU.

It was assumed that the S3 signalling channel between the retailer and the NIU had been established via management. It was also assumed that the NIU contained an appropriate number (the interactive multimedia services (IMS) equivalent of today's telephony numbers). This is transferred to the STU during the initialisation process.

The NIU also supported a local control bus for local communication with the STU.

4. Future platform development

There are a number of future development work items under discussion or ongoing for 1997/98. An overview of some of these is given in this section.

4.1 Application enhancements

Further enhancements to the end-user application are under development. One such activity recognises the role that Java and Java-compatible Web browsers have to play in the control of remote applications. Web browsers have developed from simple point-and-click interfaces to powerful operating environments in their own right, through the addition of Java processing capabilities. Future Web browsers will contain object request broker (ORB) functionality, which will enable them to perform remote method invocations on applications such as the existing DSM-CC U-U capabilities within the retailer and SPS.

With this in mind it is intended to rewrite the end-user application, including interfaces to the CORBA-based DSM-CC U-U and encapsulate it within Java. This will enable a self-contained application to be downloaded from the retailer to the end user. This application will offer the end user a GUI over which service browsing can occur through exerting control over DSM-CC service objects within the retailer or SPS. The use of DSM-CC U-U in this case is necessary for backward compatibility with the existing retailer and SPS. In the future the use of ORB-aware Java applets will enable a more generic approach to application control to be used.

4.2 Set-top box (STB) suppliers

This area of development will make use of the expertise developed to validate STB design with manufacturers. This will also include information flows and associated protocols.

Once the design has been validated and a prototype STB developed there is the potential to connect the prototype to the demonstrator platform. However, it is more likely that the expertise developed through the demonstrator platform

will be used to develop a separate validation platform for STBs. This will enable collaborating STB manufacturers to work with BT to integrate a 'telco' interface on their STBs and to validate this interface through connection to a validation platform.

4.3 Generic session control

As discussed briefly in section 2.2, the demonstrator platform has an implementation of session control based on DSM-CC U-N that is focused towards IMS, specifically video-on-demand (VoD). It has been recognised that if additional applications are to be supported, either this session control protocol must be enhanced to encompass these features or support additional session control protocols must be supported.

This area of development will look at how session control protocols, appropriate to a desired application and underlying transport network(s), could be downloaded from the network and run on the customers CPE.

5. Conclusions

The demonstrator platform has enabled the team to develop world-class expertise in call control. The results have been downstreamed, in the form of contributions, to the key standards and consensus forming bodies.

The flexible way through which the platform was developed, in terms of its architecture and use of APIs, has enabled its continued evolution to cover issues outside its original design goals. It has benefited enormously from an increase in both the visibility and applicability of the platform to related work areas.

The current demonstrator platform represents leading edge technology and has already formed the basis of a number of exhibitions.

References

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- 2 ATM-F UNI V3.0, <http://www.atmforum.com/>
- 3 ITU-T Q.2971, <http://www.itu.ch/>
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- 6 DSM-CC Specification, <ftp://playground.sun.com/pub/dsmcc/>



Paul Reece graduated from Coventry Polytechnic with a BSc (Hons) in Mathematics in 1991 and joined the Broadband and Data Networks unit.

He has been involved in a wide range of activities related to broadband signalling and network intelligence over the past six years. He has recently moved to the Intelligence Design and Performance unit where he is leading work on network intelligence and signalling architectures and design



Richard Miles joined BT in 1986 as an apprentice. He then worked on optical access network technology until 1990, when he gained sponsorship to attend the University of Kent.

He graduated in 1993 with a BEng (Hons) in Electronic Engineering and joined the Intelligence Design and Performance unit, where he has been designing and prototyping systems for the delivery of future broadband services.



Richard Macey graduated from the University of Newcastle Upon Tyne with a BEng (Hons) in Electronic Engineering in 1989 and joined a team investigating broadband networks. He has worked on the control and signalling aspects of mobile and intelligent networks, including the development of a number of demonstrator platforms. He is currently leading a task investigating the control requirements of integrated IP and ATM networks, and working in a team developing longer term network intelligence architectures.



Richard Galbraith graduated in 1995 from York University with an MEng in Electronic Systems Engineering, following a four-year sponsorship from BT. On graduation he joined the Customer Products unit working on interactive set-top boxes (STBs). He has been involved in providing technical expertise to the specification work for the British Interactive Broadcasting STB, and has also represented BT in DAVIC and the DVB. His main activity at present is working on a validation platform for broadband customer premises equipment, to validate the protocols which are required to connect STBs to future broadband networks.